

Parathyroidectomy and Blood Pressure in Hemodialysis Patients

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Key Words

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Abstract

To assess whether parathyroidectomy (PTx) affects blood pressure (BP) in hemodialysis (HD) patients, we studied 11 uremics on HD treatment for 8.2 ± 0.9 years who underwent successful PTx. As the control group, we selected 11 HD patients not submitted to PTx, matched with the study group for sex, age, years on HD, dialysis procedure and BP values. In the controls, BP and body weight did not change during the 2 years of observation. In the patients, BP remained stable in the year before PTx. PTx caused a progressive reduction in BP in 7 of the 11 patients. The fall was significant from the 3rd quarter onward (mean BP values before PTx: 139/82 mm Hg, 1 year after PTx: 122/75 mm Hg). The magnitude of the hypotensive effect of PTx was related to the **pre-PTx** systolic BP value ($r = -0.70$, $p = 0.016$). PTx also caused a significant progressive increase in body weight (1.56 ± 0.57 kg 1 year after PTx). In conclusion, PTx causes BP fall in HD patients regardless of whether their preintervention values are normal or increased. The BP reduction occurs in concomitance with a consistent increase in body weight. BP variations are clinically relevant and may be related to the **post-PTx** calcium efflux from the vessel wall.

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Introduction

The relationship between calcium (Ca), parathyroid hormone (PTH) and blood pressure (BP) is receiving increasing interest. PTH plays a permissive role on the hypertensive action of hypercalcemia [1,2], by favoring Ca entry into the smooth muscle cells of the vessel walls [1]. This effect generally prevails on the vasodilator effect that PTH plays itself [3]. Conversely, acute hypocalcemia is associated with hypotension both in humans and dogs [4, 5]. These pathophysiological studies are in keeping with the available clinical and epidemiological data. Patients with primary hyperparathyroidism show an increased prevalence of arterial hypertension, ranging from 30 to 55% [6-8]. Hypercalcemia plays a pivotal role, although not the only one [6], in sustaining the high BP values. Arterial hypertension can be cured by parathyroidectomy (PTx) in 20-55% of cases [6, 9]. Whether a similar effect exists after PTx performed for secondary hyperparathyroidism in hemodialysis (HD) patients is unknown at present. Therefore,

Table 1. Blood pressure and body weight variations

Results are means $\pm$ SE.

S = Systolic; D = diastolic; M = mean; AP = arterial pressure; c = controls; p = patients; 1, 2, 3, 4, 5, 6, 7, 8 = quarters of the 2 years of observation in c and of the year before (1-4) and after (5-8) PTx in p.

* $p < 0.05$; ** $p < 0.01$, comparisons with the last quarter before PTx by paired t test with the Bonferroni adjustment. BP and BW variations in c (quarters 1-8) and p (quarters 1-4): NS by ANOVA. In the quarters 1-4, the comparison of the means between p and c is NS by the unpaired t test.

we have retrospectively evaluated the long-term course of BP in HD patients subjected to PTx. Patients and Methods

In the period of 1980-1986, 15 uremic patients under chronic HD treatment with bone biopsy-proven overt secondary hyperparathyroidism had PTx. We have analyzed data from the 11 patients with a HD period before and after PTx each lasting at least 1 year. Of the 4 patients excluded, 2 died because of cardiovascular complications a few months after surgery and 2 underwent kidney transplantation during the follow-up.

Among our dialysis population not submitted to PTx, we selected as the controls the 11 patients better matched with the PTx patients for sex (5 men, 6 women in both groups), age (patients 49.2 ± 3.9 years, mean $\pm$ SE, controls: 49.7 ± 3.9 years), years on HD (8.2 ± 0.9 years in both groups), dialysis procedure and BP values.

In the PTx group, the pre-HD values of BP and heart rate and the post-HD values of body weight (BW) were recorded on each HD session in the year before and after the operation, so that approximately 150 values/year were obtained for each parameter analyzed for each patient. The month of PTx was excluded from the analysis. In the controls, these parameters were recorded in the same period of time.

In reviewing the clinical notes of the patients we also computed the number of episodes of symptomatic hypotension requiring treatment that occurred during each HD session both before and after surgery.

The dialysis procedure was never changed throughout the 2 years of the study. Patients and controls were dialyzed for 4 h 3 times a week with a cuprophane membrane of 1.1-1.4 m²; dialysate contained Na 138 mmol/L, K 1.5, Mg 0.75, Ca 1.75, Cl 108.5, acetate 5 and bicarbonate 31 mmol/L.

The cardiothoracic ratio was assessed on standard chest X-ray performed at least every 6 months. The immunoreactive (i) PTH was determined by RIA, using a specific antibody for the terminal carboxyl fragment. Since over the years the measurement units were changed, the results are here expressed as the ratio between the actual value and the maximum limit of normal. Serum Ca, phosphate (P) and alkaline phosphatase (Alk Phosph) measurements, performed at least every month, were carried out by standard methods.

The data were computed as the mean of the values available in each quarter, i.e., 4 before and 4 after PTx, or on a monthly basis. The results are given as mean $\pm$ SE and were analyzed by SPSS statistical package. The one-way analysis of variance (ANOVA) was used to verify the stability of BP and BW in controls and in patients before PTx. The post-PTx variations in BP and BW were analyzed by several statistics: the mean values of BP and BW in the 4 quarters after PTx

were compared with those of the last quarter before PTx by the paired t test with the Bonferroni method [10]; the percent post-PTx changes in BP were analyzed by ANOVA with the Duncan test for multiple comparisons [11]; for each patient, the post-PTx variations in BP and BW were analyzed by linear regression analysis, and the association of these parameters with time was assessed by a t test in which t is the mean of the slopes obtained for each patient divided by the SE of the mean [12]. The means between patients and controls were compared by Student's t test for unpaired data. The post PTx changes in biochemical parameters were analyzed by ANOVA with the Duncan test.

Results

Spontaneous Parameter Variations with Time In the controls, systolic, diastolic and mean arterial pressure as well as BW values did not change (ANOVA, NS) during the 2 years of observation (table 1). In the year

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before PTx, the patient BP values had remained unchanged (ANOVA, NS) and similar to those recorded in the controls (t test, NS for each of the 4 quarters analyzed; table 1). On the contrary patients lost weight, particularly in the quarter before PTx (table 1); these variations, however, were not significant by ANOVA.

Adequacy of PTx

iPTH values returned to normal within 7 days from PTx in all patients but 1 (fig. 1). In this female patient, only 3 glands were detected and excised, and iPTH, although reduced to one third of the pre-PTx values, i.e., from 11 to 4 arbitrary units, did not return to normal. One year after PTx, the iPTH values were substantially unchanged in all patients (fig. 1).

Fig. 1. iPTH values in arbitrary units before, 7 days and 1 year after PTx. Dashed area = normal range; horizontal bar = mean; ■ = values of the patient with incomplete parathyroid tissue removal (see text).

PTx-Associated Parameter Variations

By paired t test, the fall in BP and the increase in BW were significant from the 3rd quarter onwards (table 1). The analysis of BP variations with the ANOVA and Duncan test yielded similar results (fig. 2). After surgery, a significant, progressive decrease in BP took place in 7 of the 11 patients, while BW increased in all patients but 2 (table 2). The t test performed on the slopes rejected the hypothesis of no association with time, the corresponding p values being < 0.01 for both BP and BW (table 2).

The systolic BP decrease was weakly related to the increase in BW ($r = -0.27$, $p = 0.08$), but significantly related to the pre-PTx systolic values (fig. 3).

Before surgery, 2 patients were treated with hypotensive drugs; after PTx, BP reached normal values and remained stable off therapy (before PTx: BP 169/88 and 173/99 mm Hg, after PTx: 144/83 and 155/91 mm Hg, respectively). Four patients who were normotensive before surgery became hypotensive after PTx, all of them having pre-HD systolic values below 110 mm Hg.

Fig. 2. post-PTx BP variations (%) to the pre-PTx values. ANOVA: SAP and MAP $p < 0.005$, DAP $p < 0.02$. Duncan: comparison with the first quarter post-PTx, $**p < 0.05$, $*p < 0.01$.

Quarters post-PTx

Table 2. Regression of blood pressure and body weight changes over time after PTx

Heart rate remained unchanged over the study period in both patients (year before PTx: 84 ± 3 , year after PTx: 83 ± 2 beats/min) and controls (first year: 84 ± 2 , second year: 84 ± 3).

The episodes of intradialytic symptomatic hypotension increased in frequency after surgery, from 0.64 ± 0.19 to 1.43 ± 0.42 episodes/patient/treatment ($p = 0.05$ by paired t test).

Cardiothoracic ratio was 0.52 ± 0.01 before and 0.50 ± 0.01 after surgery.

As expected, serum Ca and P values dropped dramatically in the first days after the operation. Subsequently, the values increased progressively reaching the low-normal range by the end of the first month and remained so thereafter (table 3). On the contrary, Alk Phosph values gradually decreased and the changes correlated with the magnitude of systolic BP variations after PTx ($r = 0.34$, $p < 0.02$).

Before PTx, all patients received Ca supplements and 1,25-dihydroxyvitamin D₃; after surgery, the dosages were tailored on the serum Ca levels. According to our policy [13], after PTx, aluminum-containing compounds were stopped and never resumed thereafter.

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110 130 150 170

pre-PTx SAP, mm Hg

Fig. 3. Correlation between the pre-PTx systolic blood pressure values and the magnitude of post-PTx decrease: $\Delta \text{SAP post-PTx} = -0.54 \cdot \text{pre-PTx SAP} + 49.3$.

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Table 3. Post-PTx variations in serum Ca, P and Alk Phosph

Results are means $\pm$ SE.

Normal values: Ca 2.32-2.48; P 1.0-1.4; Alk Phosph 48-96. ANOVA: Ca $p < 0.05$; P and Alk Phosph $p < 0.01$.

* $p < 0.05$; ** $p < 0.01$, comparisons with the pre-PTx values; Duncan.

Discussion

The influence of PTx on BP in HD patients is poorly defined. BP reduction after PTx had been described only in anecdotal reports [14], while large reviews on the clinical effects induced by PTx do not mention BP variations [15-19]. Moreover, in studies specifically designed to define the effect of subtotal PTx on left ventricular function in HD patients, BP did not change 2 weeks after surgery [20], or mean arterial pressure (MAP) decreased by 6-10 mm Hg within 5-8 months after PTx [21]. These variations did not reach statistical significance, but it must be emphasized that, according to the study protocol, BP values were recorded only 1 day before and 1 day after surgery.

In a long-term study, we noticed that BP, measured before HD treatment 3 times a week for 1 year, progressively decreased after PTx in 64% of the patients (table 2). The fall was substantial and significant from the 3rd quarter after PTx onwards (table 1, fig. 2). The magnitude of systolic BP decrease was related to the pre-PTx values (fig. 3) and affected patients regardless of whether their presurgery values were normal or elevated.

In HD patients, the control of BP is multifactorial but we think that the pressure drop in our patients was related to PTx since BP had been constant in the year before the operation and no pressure variation was observed in the controls studied in the same period of time (table 1). It is, therefore, reasonable to exclude the possibility that the BP drop could be attributable to a natural decline with the increase of years on dialysis or to variations in the dialysis technique or to seasonal factors. Moreover, no patients displayed signs of heart failure, another potential source of hypotension.

It can be hypothesized that the BP fall is due to a reduction in Ca content in smooth muscle cells of the

arteriolar wall. To test this hypothesis, we correlated the systolic BP fall to the maximum change in Ca, P and $\text{Ca} \times \text{P}$ product before and after surgery. The lack of any significant correlation between such parameters does not argue against the hypothesis because the blood Ca and P levels are not representative of their intracellular content. On these grounds, it is tempting to speculate that the correlation between the change in systolic BP and Alk Phosph after PTx might have some relevance. The healing of bone lesions, as suggested by Alk Phosph reduction, might be temporally related to Ca efflux from the vessel wall. However, it must be remembered that a correlation does not imply a cause-effect relationship.

It is worthy of note that patients had a consistent increase in BW (tables 1, 2). Furthermore, there was a weak relationship between the changes in BW and those in systolic BP. An underestimation of patient 'dry' BW could be the underlying reason determining the BP trend after PTx, since increasing 'dry' BW does not by itself exclude under-hydration. This hypothesis cannot be easily dismissed, but we believe it to be unlikely because the cardiothoracic ratio did not change after PTx. As shown in our previous studies [22], the cardiothoracic ratio is a sensitive index of hydration status of dialysis patients since it increases with overhydration and decreases with dehydration. It follows that the post-PTx association of BW gain with no variation in cardiothoracic ratio should be interpreted as the result of improved nutritional status. In conclusion, PTx is associated with BP fall in HD patients regardless of their presurgery values. After PTx, there is a consistent BW gain probably due to increased muscle mass. The mechanism(s) underlying the hypotensive effect of PTx need(s) to be clarified by specifically designed studies.

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